

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1953

Speed 138 kV
October 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1953. The Interconnection Customer proposes a 41.2 MW solar photovoltaic facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on December 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 704 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Latah Jct to Alibates 138kV line	Lee County Solid Waste Energy to Cooper 345kV L	101.5	399	5.9
Spatsum Lornex to Latah Jct 138kV	Cooper to Lee County Solid Waste Energy 345kV	101.2	411	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Latah Jct to Alibates 138kV line	Reconductor line	\$4,272,000	1% (\$46,000)
Spatsum Lornex to Latah Jct 138kV	Reconductor line	\$2,667,000	1% (\$29,000)
	Total allocated to project		\$75,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.